TECHNICAL DILIGENCE REVIEW - RECORD 006

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-006

1. Scope of the delivery-process review and its limits

Scope of report TDR-2025-21 is the engineering delivery process, pipeline metrics and change-management controls, and the cut-off for evidence is 2025-11-06. The present report is one of a series of technical reports; this one takes the engineering delivery process, pipeline metrics and change-management controls and reports against the evidence available by 2025-11-06. Three of the open items in the present report would close on a single document that the company has agreed to produce. 4 items in the report at TDR-2025-21 are open, two of which the reviewer would treat as pre-closing information requests rather than post-close work. The evidence behind the present report is retained by Brandywine Technology Partners and is available to Ardent Peak on request.

S. Alderton was granted read access to the code, the pipeline and the incident tracker for the purposes of report TDR-2025-21; production access was not requested. Access for report TDR-2025-21 was sufficient for the conclusions drawn; where it was not, the limitation is stated at the point it bites. The pack this report proposes is deliberately short, because a longer one would be produced and not read. Brandywine Technology Partners recommends that the 5 indicators be reported with their measurement basis attached, since two of them changed basis during the reviewed period. The engineering delivery process, pipeline metrics and change-management controls is reported here at the level of detail Ardent Peak asked for and no deeper.

Two document requests remained partly unsatisfied when the present report was delivered, and the affected conclusions are marked. S. Alderton could not verify one class of claim in the report at TDR-2025-21, namely the behaviour of components deployed only at the two sites not visited. The judgement report TDR-2025-21 reaches is that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. S. Alderton would summarise the report at TDR-2025-21 in one sentence: there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. The reviewer's tone in TDR-2025-21 is deliberately flat; a finding is not made more serious by the words around it.

Coverage of this report is deliberately uneven: the safety and control path was examined closely and the reporting path only at a high level. The sample behind this report is 13 services and 5 sites; the selection was the reviewer's, not the company's. The transaction reading of report TDR-2025-21 is that the engineering risk is priced in the plan rather than in the purchase agreement. TDR-2025-21 suggests one governance change rather than a price adjustment: a quarterly technical review with the lead investor's observer present. The company's engineering leadership saw every finding in the report at TDR-2025-21 before delivery and disputed two of them.

Report TDR-2025-21 reports two findings the company disputes; the dispute is recorded next to the finding rather than resolved silently. S. Alderton has left several concerns in this report at limited evidence strength rather than dropping them or dressing them up. This report follows the reviewer's usual practice of grading evidence strength separately from finding severity, so a plausible concern is not promoted to a verified defect. S. Alderton led the work behind report TDR-2025-21, combining document review, 20 interviews and hands-on inspection of running systems. The reader of TDR-2025-21 should treat each conclusion as bounded by the section it appears in.

Interviews for the present report were conducted with the owning engineer present rather than with a manager speaking on their behalf. 15 follow-up items arose from the interview programme for the present report; 3 were still open at delivery. Brandywine Technology Partners was instructed by Ardent Peak to examine the engineering delivery process, pipeline metrics and change-management controls; the resulting report is the present report, dated 2025-11-06. The subject of report TDR-2025-21 is the engineering delivery process, pipeline metrics and change-management controls, examined for Ardent Peak by Brandywine Technology Partners over 5 weeks ending 2025-11-06. S. Alderton reports management's own position wherever it differs from the reviewer's, and does not merge the two.

Schedule U - Qualifications carried forward, with evidence strength (TDR-2025-21)

Ref	Qualification carried forward	Evidence strength	Effect on conclusion	What would close it
OQ-06-01	the remediation landed without a regression test	moderate	leaves one section unconcluded	an independent measurement

Ref	Qualification carried forward	Evidence strength	Effect on conclusion	What would close it
OQ-06-02	the remediation landed without a regression test	moderate	narrows the conclusion to the tested scale	a document the company has not produced
OQ-06-03	the objective is met on the company's own measurement only	moderate	does not change the conclusion	a document the company has not produced
OQ-06-04	the measurement basis changed during the reviewed period	limited	narrows the conclusion to the tested scale	a witnessed test
OQ-06-05	the retention window closed before the raw data was requested	moderate	leaves one section unconcluded	a decision from management
OQ-06-06	the supplier certificate was read rather than the test witnessed	moderate	does not change the conclusion	a decision from management
OQ-06-07	the supplier certificate was read rather than the test witnessed	moderate	is reflected in the post-close plan	a re-run of the original test
OQ-06-08	the second bench rig does not cover this path	strong	narrows the conclusion to the tested scale	a re-run of the original test
OQ-06-09	two teams each believe the other owns this	strong	is reflected in the post-close plan	a re-run of the original test
OQ-06-10	the population completeness could not be proven	limited	leaves one section unconcluded	a witnessed test
OQ-06-11	the declined item has no expiry date	moderate	is reflected in the post-close plan	a decision from management
OQ-06-12	the second bench rig does not cover this path	moderate	narrows the conclusion to the tested scale	an independent measurement
OQ-06-13	the objective is met on the company's own measurement only	strong	does not change the conclusion	a document the company has not produced
OQ-06-14	the second bench rig does not cover this path	strong	narrows the conclusion to the tested scale	a re-run of the original test
OQ-06-15	the retention window closed before the raw data was requested	strong	would change the conclusion if resolved adversely	an independent measurement
OQ-06-16	the second bench rig does not cover this path	limited	does not change the conclusion	a re-run of the original test
OQ-06-17	the remediation landed without a regression test	strong	is reflected in the post-close plan	a witnessed test
OQ-06-18	two teams each believe the other owns this	moderate	does not change the conclusion	one further interview
OQ-06-19	the second bench rig does not cover this path	strong	would change the conclusion if resolved adversely	one further interview
OQ-06-20	the supplier certificate was read rather than the test witnessed	strong	is reflected in the post-close plan	an independent measurement
OQ-06-21	the supplier certificate was read rather than the test witnessed	strong	would change the conclusion if resolved adversely	a decision from management
OQ-06-22	the compensating control was not tested	strong	leaves one section unconcluded	an independent measurement
OQ-06-23	the population completeness could not be proven	strong	is reflected in the post-close plan	a witnessed test
OQ-06-24	the objective is met on the company's own measurement only	strong	would change the conclusion if resolved adversely	an independent measurement
OQ-06-25	the declined item has no expiry date	strong	leaves one section unconcluded	one further interview
OQ-06-26	the measurement basis changed during the reviewed period	moderate	narrows the conclusion to the tested scale	one further interview
OQ-06-27	the retention window closed before the raw data was requested	moderate	would change the conclusion if resolved adversely	a re-run of the original test
OQ-06-28	the declined item has no expiry date	limited	narrows the conclusion to the tested scale	a witnessed test

Ref	Qualification carried forward	Evidence strength	Effect on conclusion	What would close it
OQ-06-29	the measurement basis changed during the reviewed period	moderate	is reflected in the post-close plan	a document the company has not produced
OQ-06-30	the remediation landed without a regression test	moderate	would change the conclusion if resolved adversely	an independent measurement
OQ-06-31	the measurement basis changed during the reviewed period	limited	leaves one section unconcluded	an independent measurement
OQ-06-32	the supplier certificate was read rather than the test witnessed	strong	narrows the conclusion to the tested scale	one further interview

2. Pipeline stages, durations and gate configuration

Timing on unit tests, cloud services is 6 m 20 s mean, 9 m 05 s at p90, measured over the reviewed quarter. Unit tests, cloud services carries this qualification: two suites account for most of the flakiness, both involving a shared database fixture. In the merge path unit tests, cloud services is blocking. unit tests, cloud services has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Unit tests, robot software shows 7.8% failures and 4.6% flakes, and the reviewer confirmed the split is measured rather than estimated. Blocking is the gate configuration for unit tests, robot software. Unit tests, robot software is maintained by Autonomy Core. unit tests, robot software was raised again with the owning engineer after the initial interview, and the position did not change.

Autonomy Core maintains static analysis, robot software and answers for its reliability. A share that is invisible because it runs in parallel is what static analysis, robot software adds to the time between merge and production. Over the reviewed period the static analysis, robot software stage degraded after a new rule set was added. static analysis, robot software was one of two items where TDR-2025-21 could not obtain the underlying data within the engagement.

Build, cloud container images carries a 2.7% failure rate; the flake component is 0.3%. Build, cloud container images was stable in every respect the reviewer could measure, and the team could explain why. Maintenance of build, cloud container images falls to Platform Infrastructure. build, cloud container images was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Build, robot firmware images carries a 4.1% failure rate; the flake component is 0.6%. Reading the build, robot firmware images metrics against the merge policy, the cross toolchain is rebuilt weekly, which produces a periodic slow build. 19 m 30 s is the mean duration of build, robot firmware images, and 31 m 20 s is the figure the reviewer would plan against. build, robot firmware images was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Integration tests, cloud is responsible for a small and predictable share of the total pipeline time. The failure rate on integration tests, cloud is 12.6%, with 8.2% attributable to flakiness. On integration tests, cloud the reviewer records that the highest flake rate in the pipeline and the most common cause of a retried run. integration tests, cloud carries a follow-up request that was satisfied after the draft of TDR-2025-21 was circulated.

Contract tests, warehouse connectors costs 9 m 40 s typically and 15 m 55 s when the build fleet is contended. Contract tests, warehouse connectors sits in the pipeline as blocking, and the reviewer verified the configuration rather than the documentation. Contract tests, warehouse connectors shows 9.3% failures and 3.7% flakes, and the reviewer confirmed the split is measured rather than estimated. contract tests, warehouse connectors is reported at the evidence strength available and has not been promoted to fit the narrative.

Simulation suite, short was stable in every respect the reviewer could measure across the measured window. The lead-time cost of simulation suite, short is little on a clean run and a great deal on a contended one. In the merge path simulation suite, short is blocking. simulation suite, short rests on a single source, and the reviewer has said so rather than implying corroboration.

Schedule Q - Pipeline stage duration, failure rate and gate configuration (TDR-2025-21)

Pipeline stage	Mean duration	p90 duration	Failure rate	Flaky rate	Gate	Owning team
lint and format	1 m 10 s	1 m 40 s	3.1%	0.1%	blocking	Developer Productivity
unit tests, cloud services	6 m 20 s	9 m 05 s	5.4%	1.9%	blocking	Fleet Services
unit tests, robot software	11 m 45 s	18 m 30 s	7.8%	4.6%	blocking	Autonomy Core
static analysis, robot software	14 m 05 s	22 m 10 s	6.2%	0.4%	blocking	Autonomy Core
build, cloud container images	8 m 15 s	12 m 40 s	2.7%	0.3%	blocking	Platform Infrastructure
build, robot firmware images	19 m 30 s	31 m 20 s	4.1%	0.6%	blocking	Robot Firmware
integration tests, cloud	24 m 50 s	41 m 15 s	12.6%	8.2%	blocking	Fleet Services
contract tests, warehouse connectors	9 m 40 s	15 m 55 s	9.3%	3.7%	blocking	Integrations
simulation suite, short	33 m 10 s	52 m 00 s	8.9%	5.1%	blocking	Autonomy Core
simulation suite, nightly	3 h 40 m	4 h 25 m	14.2%	6.8%	advisory	Autonomy Core
hardware-in-the-loop, bench	1 h 15 m	2 h 05 m	10.4%	2.2%	blocking for firmware	Robot Firmware
dependency and licence scan	4 m 25 s	6 m 50 s	1.8%	0.2%	advisory	Security Engineering
secret scan	2 m 05 s	3 m 15 s	0.9%	0.1%	blocking	Security Engineering
container image vulnerability scan	5 m 35 s	8 m 20 s	16.7%	1.1%	advisory	Security Engineering
performance smoke, cloud	12 m 20 s	19 m 45 s	6.6%	3.4%	advisory	Platform Infrastructure
release packaging and signing	7 m 50 s	11 m 10 s	2.2%	0.2%	blocking	Developer Productivity

3. Failure and flake rates, and what they cost the team

Simulation suite, nightly takes 3 h 40 m on average and 4 h 25 m at the 90th percentile. The reviewer's observation on simulation suite, nightly is that advisory only, which means a nightly regression can reach a release candidate. Whether simulation suite, nightly blocks matters: it is advisory. simulation suite, nightly was cross-checked against the engineering record supplied for TDR-2025-21, and the two agree.

For hardware-in-the-loop, bench the pipeline reports 1 h 15 m mean and 2 h 05 m at the 90th percentile. Hardware-in-the-loop, bench was stable in every respect the reviewer could measure, and the team could explain why. The reviewer's observation on hardware-in-the-loop, bench is that queueing on two rigs is the main contributor to firmware lead time. hardware-in-the-loop, bench was raised again with the owning engineer after the initial interview, and the position did not change.

Over the reviewed period the dependency and licence scan stage improved in flakiness after two suites were quarantined. Dependency and licence scan takes 4 m 25 s on average and 6 m 50 s at the 90th percentile. Ownership of dependency and licence scan sits with Security Engineering. dependency and licence scan is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Secret scan improved after the build fleet was enlarged across the measured window. Secret scan contributes more than the two preceding stages combined to the change lead time. Duration of secret scan runs 2 m 05 s to 3 m 15 s across the measured window. secret scan is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Timing on container image vulnerability scan is 5 m 35 s mean, 8 m 20 s at p90, measured over the reviewed quarter. Container image vulnerability scan carries this qualification: the failure rate reflects new advisories rather than new defects. Container image vulnerability scan is responsible for the largest single share of any stage of the total pipeline time. container image vulnerability scan was one of two items where TDR-2025-21 could not obtain the underlying data within the engagement.

About performance smoke, cloud the reviewer would add that thresholds were last recalibrated in 2024 and are now loose. In lead-time terms performance smoke, cloud contributes a share that is invisible because it runs in parallel. The failure rate on performance smoke, cloud is 6.6%, with 3.4% attributable to flakiness. performance smoke, cloud is the entry that changed most between the first and second document requests behind TDR-2025-21.

Over the reviewed period the release packaging and signing stage was stable in every respect the reviewer could measure. Release packaging and signing takes 7 m 50 s on average and 11 m 10 s at the 90th percentile. Of the end-to-end lead time, release packaging and signing accounts for a small and predictable share. release packaging and signing was raised again with the owning engineer after the initial interview, and the position did not change.

Lint and format is blocking in the merge path. Lint and format carries a 3.1% failure rate; the flake component is 0.1%. Lint and format is maintained by Developer Productivity. lint and format is treated as material to the investment case rather than as a routine engineering item.

Schedule R - Release history sampled across the reviewed period (TDR-2025-21)

Release	Date	Component	Gate set applied	Approver	Rollback	Outcome
R6.0-101	2025-08-01	Battery Management Service	full gate set	Marta Kellerman	rolled back same day	promoted to ring one only
R7.8-102	2025-08-03	Gripper Controller	full gate set	Marcus Duplessis	rolled back next day	promoted to all rings
R6.9-103	2025-08-09	Costmap Fusion	full gate set	Marta Kellerman	rolled back next day	paused at ring two
R6.8-104	2025-08-13	Charge Scheduler	emergency path	Owen Brantley	not required	promoted to all rings
R6.1-105	2025-08-15	Map Authoring Studio	full gate set	Owen Brantley	not required	withdrawn
R8.5-106	2025-08-21	E-Stop Chain	full gate set	Peter Halvorsen	not required	promoted to ring one only
R6.0-107	2025-08-25	Fleet Health Monitor	gate set with the bench suite waived	Farida Cisse	not required	promoted to all rings
R6.12-108	2025-08-29	Telemetry Ingest	full gate set	Farida Cisse	not required	paused at ring two
R7.1-109	2025-09-01	Release Orchestrator	gate set with the bench suite waived	Anjali Sekhon	rolled back next day	paused at ring two
R5.0-110	2025-09-06	WMS Connector	gate set with the nightly suite waived	Colin Achebe	rolled back next day	paused at ring two
R8.0-111	2025-09-09	Costmap Fusion	full gate set	Bettina Vogel	not required	promoted to all rings
R8.8-112	2025-09-14	WMS Connector	gate set with the bench suite waived	Bettina Vogel	rolled back next day	promoted to all rings
R4.5-113	2025-09-16	Fleet Allocator	full gate set	Anjali Sekhon	not required	withdrawn
R6.0-114	2025-09-21	Robot Shadow Store	full gate set	Nadia Semenova	not required	withdrawn
R8.0-115	2025-09-26	Telemetry Ingest	gate set with the nightly suite waived	Nathan Oduya	not required	promoted to all rings
R7.3-116	2025-09-30	Robot Shadow Store	full gate set	Bettina Vogel	not required	paused at ring two
R7.6-117	2025-10-03	E-Stop Chain	gate set with the bench suite waived	Marta Kellerman	not required	promoted to ring one only
R8.7-118	2025-10-07	Billing Usage Meter	gate set with the nightly suite waived	Anjali Sekhon	not required	withdrawn
R5.2-119	2025-10-10	Billing Usage Meter	gate set with the nightly suite waived	Sam Oyelaran	not required	promoted to all rings
R4.5-120	2025-10-16	Billing Usage Meter	full gate set	Sonia Krall	not required	promoted to all rings
R4.4-121	2025-10-19	Analytics Warehouse Loader	full gate set	Lucia Ferrante	not required	promoted to ring one only
R4.11-122	2025-10-23	E-Stop Chain	full gate set	Yuki Harada	not required	promoted to all rings

Release	Date	Component	Gate set applied	Approver	Rollback	Outcome
R4.4-123	2025-10-27	Edge Gateway Agent	full gate set	Ines Duarte	not required	promoted to all rings
R7.6-124	2025-10-30	Fleet Allocator	gate set with the bench suite waived	Bettina Vogel	not required	promoted to all rings
R5.6-125	2025-11-03	Site Provisioning Service	full gate set	Rahul Menon	rolled back next day	promoted to all rings
R6.12-126	2025-11-07	WMS Connector	gate set with the nightly suite waived	Sam Oyelaran	not required	promoted to all rings
R8.13-127	2025-11-12	Edge Gateway Agent	full gate set	Wanda Njoku	rolled back next day	withdrawn
R4.10-128	2025-11-17	Analytics Warehouse Loader	full gate set	Bettina Vogel	not required	promoted to all rings
R4.3-129	2025-11-19	Fleet Health Monitor	full gate set	Yuki Harada	not required	promoted to all rings
R5.8-130	2025-11-25	Analytics Warehouse Loader	emergency path	Wanda Njoku	not required	promoted to ring one only
R8.3-131	2025-11-28	Order Broker	gate set with the nightly suite waived	Bettina Vogel	not required	paused at ring two
R5.2-132	2025-12-02	Localization Engine	gate set with the bench suite waived	Erik Palmgren	rolled back next day	promoted to all rings
R6.9-133	2025-12-06	Simulation Harness	full gate set	Marta Kellerman	not required	promoted to ring one only
R4.7-134	2025-12-11	Traffic Manager	gate set with the bench suite waived	Erik Palmgren	not required	withdrawn
R8.6-135	2025-12-15	Hardware-in-the-Loop Rig	gate set with the bench suite waived	Lucia Ferrante	not required	promoted to all rings
R4.9-136	2025-12-17	Telemetry Ingest	full gate set	Rahul Menon	rolled back next day	promoted to all rings
R8.2-137	2025-12-21	Safety Controller	full gate set	Daniel Okafor	rolled back next day	promoted to all rings
R5.0-138	2025-12-27	Release Orchestrator	full gate set	Wanda Njoku	rolled back next day	withdrawn
R8.5-139	2025-12-31	Site Provisioning Service	full gate set	Sonia Krall	rolled back same day	promoted to all rings
R5.0-140	2026-01-03	Fleet Allocator	gate set with the nightly suite waived	Ines Duarte	rolled back next day	promoted to all rings
R7.5-141	2026-01-06	WMS Connector	emergency path	Sam Oyelaran	not required	promoted to ring one only
R6.8-142	2026-01-11	Robot Shadow Store	full gate set	Erik Palmgren	not required	paused at ring two
R8.1-143	2026-01-15	Release Orchestrator	gate set with the nightly suite waived	Owen Brantley	rolled back same day	promoted to ring one only
R4.0-144	2026-01-18	Traffic Manager	full gate set	Grace Lindqvist	not required	promoted to ring one only
R6.4-145	2026-01-22	Billing Usage Meter	gate set with the nightly suite waived	Erik Palmgren	rolled back next day	promoted to ring one only
R7.10-146	2026-01-28	Robot Shadow Store	gate set with the bench suite waived	Marta Kellerman	not required	paused at ring two
R8.13-147	2026-02-01	Analytics Warehouse Loader	gate set with the nightly suite waived	Nadia Semenova	not required	paused at ring two
R6.0-148	2026-02-04	Robot Shadow Store	full gate set	Owen Brantley	rolled back next day	paused at ring two

3. Failure and flake rates, and what they cost the team - part 2 of 2

Presentation of the 3. Failure and flake rates, and what they cost the team carries on below without restarting the row numbers, the break being one of pagination only. Pagination alone determines where it falls.

Definitions given once for the 3. Failure and flake rates, and what they cost the team are relied on again rather than restated, the compilation basis being identical. The parts are to be read together.

Schedule R - Release history sampled across the reviewed period (TDR-2025-21) (continued, part 2 of 2)

Release	Date	Component	Gate set applied	Approver	Rollback	Outcome
R8.13-149	2026-02-07	Billing Usage Meter	gate set with the bench suite waived	Marta Kellerman	not required	withdrawn
R5.7-150	2026-02-12	WMS Connector	gate set with the bench suite waived	Nathan Oduya	not required	promoted to all rings
R8.11-151	2026-02-17	Config Distribution	full gate set	Erik Palmgren	not required	promoted to all rings
R7.12-152	2026-02-20	Event Bus	full gate set	Tomas Villareal	rolled back same day	paused at ring two
R8.0-153	2026-02-25	WMS Connector	gate set with the bench suite waived	Marta Kellerman	not required	promoted to ring one only
R7.5-154	2026-03-01	Gripper Controller	emergency path	Nadia Semenova	not required	promoted to all rings
R8.13-155	2026-03-03	Motion Planner	full gate set	Grace Lindqvist	not required	promoted to ring one only
R4.4-156	2026-03-08	Charge Scheduler	gate set with the nightly suite waived	Nadia Semenova	rolled back next day	withdrawn
R5.0-157	2026-03-12	Identity and Authorization	full gate set	Daniel Okafor	not required	withdrawn
R6.0-158	2026-03-15	Costmap Fusion	emergency path	Erik Palmgren	not required	paused at ring two
R7.6-159	2026-03-20	Safety Controller	gate set with the nightly suite waived	Yuki Harada	rolled back same day	promoted to all rings
R4.10-160	2026-03-24	Analytics Warehouse Loader	gate set with the nightly suite waived	Marta Kellerman	rolled back next day	paused at ring two
R7.5-161	2026-03-27	Billing Usage Meter	emergency path	Wanda Njoku	not required	promoted to all rings
R5.11-162	2026-04-02	Simulation Harness	emergency path	Priya Raghunathan	not required	withdrawn
R5.10-163	2026-04-05	Speed and Separation Monitor	gate set with the bench suite waived	Sam Oyelaran	rolled back same day	paused at ring two
R8.8-164	2026-04-08	Charge Scheduler	gate set with the bench suite waived	Sonia Krall	rolled back next day	paused at ring two
R6.2-165	2026-04-13	Speed and Separation Monitor	full gate set	Farida Cisse	not required	withdrawn
R8.9-166	2026-04-18	E-Stop Chain	gate set with the nightly suite waived	Grace Lindqvist	not required	withdrawn
R8.6-167	2026-04-22	Motion Planner	full gate set	Nadia Semenova	not required	withdrawn
R5.11-168	2026-04-25	Identity and Authorization	gate set with the nightly suite waived	Wanda Njoku	not required	withdrawn
R8.7-169	2026-04-28	Release Orchestrator	gate set with the nightly suite waived	Nathan Oduya	not required	promoted to all rings
R5.11-170	2026-05-04	Battery Management Service	full gate set	Sam Oyelaran	not required	promoted to all rings
R7.6-171	2026-05-07	Edge Gateway Agent	full gate set	Tomas Villareal	rolled back same day	promoted to all rings
R6.5-172	2026-05-11	Event Bus	gate set with the nightly suite waived	Theo Bassett	not required	promoted to all rings
R4.11-173	2026-05-15	Alarm Router	full gate set	Theo Bassett	not required	paused at ring two
R5.3-174	2026-05-19	Billing Usage Meter	gate set with the bench suite waived	Colin Achebe	rolled back same day	promoted to all rings
R8.13-175	2026-05-24	Identity and	full gate set	Marta Kellerman	rolled back same	promoted to all

Release	Date	Component	Gate set applied	Approver	Rollback	Outcome
		Authorization			day	rings
R7.0-176	2026-05-28	Motion Planner	full gate set	Nadia Semenova	rolled back same day	promoted to all rings
R5.11-177	2026-06-01	Hardware-in-the-Loop Rig	full gate set	Anjali Sekhon	rolled back same day	promoted to all rings
R6.9-178	2026-06-03	Map Authoring Studio	full gate set	Erik Palmgren	not required	paused at ring two
R8.5-179	2026-06-09	Robot Shadow Store	gate set with the bench suite waived	Marcus Duplessis	rolled back same day	withdrawn
R5.10-180	2026-06-11	Event Bus	gate set with the bench suite waived	Marta Kellerman	rolled back next day	paused at ring two
R5.1-181	2026-06-15	Analytics Warehouse Loader	gate set with the bench suite waived	Rahul Menon	rolled back same day	paused at ring two
R5.7-182	2026-06-20	Gripper Controller	full gate set	Nathan Oduya	rolled back next day	promoted to all rings
R7.8-183	2026-06-25	Edge Gateway Agent	full gate set	Farida Cisse	rolled back next day	promoted to all rings
R4.4-184	2026-06-27	Remote Operations Console	gate set with the bench suite waived	Nathan Oduya	not required	promoted to ring one only
R7.9-185	2026-07-01	Site Provisioning Service	full gate set	Bettina Vogel	not required	promoted to all rings
R4.8-186	2026-07-07	Telemetry Ingest	full gate set	Yuki Harada	not required	promoted to all rings
R5.12-187	2026-07-10	Edge Gateway Agent	full gate set	Peter Halvorsen	not required	withdrawn
R6.8-188	2026-07-14	Alarm Router	full gate set	Colin Achebe	not required	withdrawn
R5.9-189	2026-07-17	Gripper Controller	full gate set	Farida Cisse	not required	promoted to all rings
R6.5-190	2026-07-22	WMS Connector	gate set with the nightly suite waived	Colin Achebe	not required	promoted to ring one only
R4.10-191	2026-07-27	Motion Planner	full gate set	Anjali Sekhon	not required	paused at ring two
R7.7-192	2026-07-29	Hardware-in-the-Loop Rig	gate set with the bench suite waived	Sam Oyelaran	not required	promoted to all rings
R4.13-193	2026-08-03	Speed and Separation Monitor	full gate set	Farida Cisse	not required	promoted to all rings
R7.13-194	2026-08-08	Site Provisioning Service	gate set with the nightly suite waived	Wanda Njoku	not required	promoted to all rings
R5.8-195	2026-08-10	Localization Engine	emergency path	Rahul Menon	not required	paused at ring two
R6.6-196	2026-08-16	Edge Gateway Agent	full gate set	Colin Achebe	not required	paused at ring two

4. Release cadence and ownership across the estate

Platform Infrastructure releases Event Bus upgraded twice a year, and no release in the sampled window bypassed the gate set. Against Event Bus the backlog runs to 4 items, and it includes a known behaviour whereby one topic carries both control and telemetry traffic, so a telemetry backlog delays control events. Against Event Bus, the coverage figure is no meaningful automated unit coverage, verification being by witnessed test instead, which coverage alone overstates the assurance here, because the bench suite is the real safety net. Placement of Event Bus is cloud, one cluster per region, three availability zones, which the reviewer tested by tracing a request end to end. Reading the dependency map from Event Bus outward gives every publishing service on the input side and every consuming service on the output side. Event Bus is the clearest instance in TDR-2025-21 of a control that exists on paper and is weaker in practice.

End-of-life review of Event Bus found the broker minor release in production leaves the vendor's support window in 2027. Event Bus was checked for shared-database coupling, and the ownership boundary held. The objective recorded against Event Bus is that publish acknowledgement inside 25 ms at p95 and no partition without a leader, and it is reported weekly to the engineering leadership. What the reviewer had for Event Bus was the cluster configuration export, topic inventory and a partition failover exercise; the conclusions below do not reach past it. Read against its objective,

Event Bus management's own assessment is broadly supported, with the coverage gap treated as a genuine finding rather than a presentational one. Event Bus appears twice in the working papers for TDR-2025-21 because the first answer was corrected after a follow-up request.

Order Broker has a single named owner, Grace Lindqvist of Fleet Services, which is not true of every service reviewed. For Order Broker the supportability position is none identified. Order Broker pairs a Go 1.22 implementation with cloud, containers on a managed Kubernetes cluster, and its data lives in PostgreSQL 14 with an event journal. Against Order Broker the backlog runs to 11 items, and it includes a known behaviour whereby order cancellation races with assignment and can leave an orphaned task record. Order Broker is deployed cloud, regional. Order Broker is the entry that changed most between the first and second document requests behind TDR-2025-21.

On Order Broker the conclusion is qualified: it is fit for the current fleet size, and the reviewer found no evidence that it would remain so at twice that scale. Order Broker ships weekly; the reviewer sampled the last twelve releases. Inputs to Order Broker come from the WMS connector and the site configuration service, and its consumers are the fleet allocator and the analytics warehouse loader. For Order Broker the stated target is that no accepted order is lost, measured by journal reconciliation each night, expressed against production traffic rather than a benchmark. 78 per cent line coverage is reported for Order Broker, and the reviewer re-ran the measurement on a clean checkout. Order Broker was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Alarm Router is deployed cloud, regional, with a second delivery provider. 51 per cent line coverage is reported for Alarm Router, and the reviewer re-ran the measurement on a clean checkout. The cadence recorded for Alarm Router is fortnightly, which matches the observed release history. Reading the dependency map from Alarm Router outward gives the fleet health monitor and the on-call rotation source on the input side and the paging provider and the customer notification channel on the output side. The published objective for Alarm Router is that a dispatched alarm reaches an on-call responder inside 90 s at p95. Alarm Router would move from a concern to a verified defect if one further measurement were supplied.

Alarm Router carries 14 open defects, among them a case where an on-call rotation gap in the routing table falls back to a shared mailbox nobody watches at night. Alarm Router was assessed on the routing table export, delivery receipts for one quarter and two missed-page reviews, and nothing beyond that was available in the data room. Alarm Router was checked for shared-database coupling, and the ownership boundary held. Alarm Router is owned by the Field Reliability team, with Wanda Njoku named as the engineering owner of record. End-of-life review of Alarm Router found the paging provider's older application interface is still in use. Alarm Router was checked against the corresponding entry in the company's own register, and the register was current.

Identity and Authorization is owned by the Security Engineering team, with Sonia Krall named as the engineering owner of record. On end-of-life exposure, Identity and Authorization shows none identified in the reviewed manifests. Identity and Authorization ships weekly, with a security review on every change; the reviewer sampled the last twelve releases. Identity and Authorization carries a single measurable objective, namely that token issuance inside 200 ms at p95 and no unauthenticated write path. Identity and Authorization sits at 84 per cent line coverage; the trend across the reviewed period was up by three points. Identity and Authorization was reviewed against the position twelve months earlier, and the direction of travel is favourable.

The assessment of Identity and Authorization is not unqualified: the objective is met in production, but the evidence that it will be met after the next dependency upgrade is weak. Identity and Authorization is deployed cloud, global control plane with regional caches. Identity and Authorization carries 6 open defects, among them a case where service-to-service tokens are issued with a twelve-hour lifetime, longer than the design intent. Evidence examined for Identity and Authorization comprised the authorization model document, a token lifetime audit and the secret rotation log. Identity and Authorization owns its own data and exposes it only through a versioned interface, which the reviewer read. Identity and Authorization is the only entry in its group where the reviewer and the engineering team reached different conclusions.

The backlog for Traffic Manager is 12 items deep and contains an acknowledged defect whereby a corridor reservation leaks when a robot drops off the network mid-traverse. Traffic Manager was assessed on the reservation protocol specification and a replay of three deadlock events, and nothing beyond that was available in the data room. Taking the evidence together, Traffic Manager it is fit for the current fleet size, and the reviewer found no evidence that it would remain so at twice that scale. Traffic Manager does not share a schema with any consumer; every read by the

motion planner and the remote operations console goes through the published interface. The topology for Traffic Manager is cloud, regional, with a degraded on-robot yield behaviour if the link drops, and the reviewer confirmed it against the deployment manifests. Traffic Manager is treated as material to the investment case rather than as a routine engineering item.

On paper Traffic Manager releases weekly, paired with the allocator release, and the observed practice did not diverge from that. Traffic Manager was built in Go 1.22 and has not been rewritten; it still runs on cloud, containers on a managed Kubernetes cluster against Redis 7 for corridor reservations with a PostgreSQL 14 audit trail. Against Traffic Manager, the coverage figure is 74 per cent line coverage, which the reviewer notes that the untested share includes the error handling. The objective recorded against Traffic Manager is that corridor deadlock detected and broken inside two seconds, and it is reported weekly to the engineering leadership. Ownership of Traffic Manager sits with Fleet Services; Owen Brantley signs off changes to it. Traffic Manager carries a follow-up request that was satisfied after the draft of TDR-2025-21 was circulated.

Wanda Njoku of Field Reliability maintains Fleet Health Monitor, and the service inventory entry was current at the review date. End-of-life review of Fleet Health Monitor found none identified. Taking the evidence together, Fleet Health Monitor the position is stronger than the coverage figure alone would suggest, because the bench evidence is unusually good here. Deployment of Fleet Health Monitor is cloud, regional, which sets the blast radius of any defect in it. Against Fleet Health Monitor, the coverage figure is 57 per cent line coverage, which the figure is respectable but excludes the fallback path, which is where the known defect sits. Fleet Health Monitor appears twice in the working papers for TDR-2025-21 because the first answer was corrected after a follow-up request.

The backlog for Fleet Health Monitor is 17 items deep and contains an acknowledged defect whereby alert grouping collapses distinct faults at the same site into one case, hiding the second fault. A time-series store plus PostgreSQL 14 for case records is the system of record for Fleet Health Monitor, a Python 3.11 component running on cloud, containers on a managed Kubernetes cluster. Service-level intent for Fleet Health Monitor states that a fleet-affecting condition raises an alert inside three minutes of first symptom. On paper Fleet Health Monitor releases fortnightly, and the observed practice did not diverge from that. For Fleet Health Monitor the reviewer worked from the alert rule inventory, three months of case records and a rule coverage review. Fleet Health Monitor was one of two items where TDR-2025-21 could not obtain the underlying data within the engagement.

Costmap Fusion sits at 58 per cent line coverage; the trend across the reviewed period was up by four points. The motion planner and the speed and separation monitor rely on Costmap Fusion, which in turn relies on the lidar driver, the depth camera bridge and the wheel odometry filter. One coupling concern applies to Costmap Fusion: a shared-memory ring buffer with no durable persistence is shared with a second service, so a schema change is not a local decision. Against Costmap Fusion the backlog runs to 21 items, and it includes a known behaviour whereby a stale lidar frame is retained for one extra cycle while a driver reconnects. Costmap Fusion was assessed on the fusion design note, sensor timing captures and the reviewer's own bench replay, and nothing beyond that was available in the data room. Costmap Fusion is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

The reviewer's judgement on Costmap Fusion is that the component is adequate today and carries a supportability liability that is not on the funded plan. End-of-life review of Costmap Fusion found a vendored geometry library whose upstream branch left support in 2024. Release cadence for Costmap Fusion is monthly, batched with perception model updates, and the release records bear that out. Daniel Okafor leads Perception's work on Costmap Fusion and was the reviewer's principal interview for it. Costmap Fusion sits edge, co-scheduled with the planner on the compute module, and no alternative placement is currently supported. Costmap Fusion was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Automated test coverage on Gripper Controller stands at 41 per cent line coverage. On Gripper Controller the conclusion is qualified: there is no finding of unfitness; there is a finding that the team cannot currently demonstrate fitness. Ines Duarte of Robot Firmware maintains Gripper Controller, and the service inventory entry was current at the review date. For Gripper Controller the supportability position is none; the coprocessor part is current production. Gripper Controller sits between the payload scale interface and the perception grasp proposals and the order execution path and the safety controller fault input, and that position is what makes its availability objective load-bearing. Gripper Controller was one of two items where TDR-2025-21 could not obtain the underlying data within the engagement.

Gripper Controller was assessed on the gripper control specification, calibration records for eleven units and a force bench test, and nothing beyond that was available in the data room. The published objective for Gripper Controller is that a grasp completes or reports failure inside 1.2 s. Robot Firmware tracks 15 open defects on Gripper Controller, including the acknowledged case where force limiting relies on a calibration that is only re-run when a gripper is replaced. One coupling concern applies to Gripper Controller: a local parameter store is shared with a second service, so a schema change is not a local decision. Deployment of Gripper Controller is edge, only on the two robot variants fitted with a manipulator, which sets the blast radius of any defect in it. Gripper Controller is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Schedule G - Open defect population and ageing by service (TDR-2025-21)

Service	Open defects	Oldest open item	Severity mix	Closed in reviewed period	Engineering owner
Motion Planner	14	2025-06-17	0 high, 4 medium, 10 low	4	Priya Raghunathan
Costmap Fusion	21	2025-03-28	2 high, 4 medium, 15 low	14	Daniel Okafor
Fleet Allocator	9	2025-10-21	2 high, 4 medium, 3 low	5	Marta Kellerman
Traffic Manager	12	2025-07-21	2 high, 3 medium, 7 low	3	Owen Brantley
Charge Scheduler	7	2025-06-16	2 high, 1 medium, 4 low	5	Nadia Semenova
Safety Controller	3	2025-03-14	2 high, 2 medium, 0 low	11	Bettina Vogel
E-Stop Chain	2	2025-08-01	2 high, 1 medium, 0 low	11	Theo Bassett
Lidar Driver	18	2025-03-21	1 high, 3 medium, 14 low	3	Yuki Harada
Localization Engine	16	2025-12-20	1 high, 8 medium, 7 low	11	Rahul Menon
WMS Connector	23	2025-12-01	2 high, 8 medium, 13 low	2	Tomas Villareal
Order Broker	11	2025-03-24	1 high, 2 medium, 8 low	6	Grace Lindqvist
Telemetry Ingest	19	2025-11-07	2 high, 7 medium, 10 low	2	Farida Cisse
OTA Updater	8	2025-12-05	2 high, 4 medium, 2 low	4	Erik Palmgren
Identity and Authorization	6	2026-02-16	2 high, 3 medium, 1 low	7	Sonia Krall
Remote Operations Console	27	2026-01-16	0 high, 8 medium, 19 low	13	Anjali Sekhon
Simulation Harness	31	2025-10-25	1 high, 2 medium, 28 low	2	Colin Achebe
Hardware-in-the-Loop Rig	13	2026-01-29	0 high, 3 medium, 10 low	6	Nathan Oduya

4. Release cadence and ownership across the estate - part 2 of 2

Continuing the 4. Release cadence and ownership across the estate, the rows below follow directly on from part 2 minus one, with no gap in the reference sequence. The sequence is unbroken either side of it.

No basis of compilation changes at this break; whatever the 4. Release cadence and ownership across the estate recorded about scope and method at its head continues to apply. The sequence is unbroken either side of it.

Schedule G - Open defect population and ageing by service (TDR-2025-21) (continued, part 2 of 2)

Service	Open defects	Oldest open item	Severity mix	Closed in reviewed period	Engineering owner
Fleet Health Monitor	17	2026-04-05	2 high, 5 medium, 10 low	13	Wanda Njoku
Battery Management Service	10	2025-04-23	2 high, 2 medium, 6 low	9	Peter Halvorsen
Gripper Controller	15	2025-03-29	1 high, 2 medium, 12 low	1	Ines Duarte
Map Authoring Studio	22	2025-07-04	0 high, 5 medium, 17 low	14	Lucia Ferrante

Service	Open defects	Oldest open item	Severity mix	Closed in reviewed period	Engineering owner
Site Provisioning Service	5	2025-06-12	1 high, 2 medium, 2 low	2	Marcus Duplessis
Event Bus	4	2025-07-30	0 high, 2 medium, 2 low	10	Erik Palmgren
Config Distribution	6	2026-04-05	1 high, 1 medium, 4 low	3	Sam Oyelaran
Robot Shadow Store	9	2025-12-05	1 high, 2 medium, 6 low	10	Marta Kellerman
Analytics Warehouse Loader	20	2026-04-16	2 high, 2 medium, 16 low	7	Farida Cisse
Billing Usage Meter	7	2025-04-28	2 high, 2 medium, 3 low	13	Nadia Semenova
Alarm Router	14	2025-10-16	2 high, 5 medium, 7 low	13	Wanda Njoku
Video Snippet Service	12	2025-08-15	0 high, 2 medium, 10 low	1	Anjali Sekhon
Speed and Separation Monitor	4	2025-06-30	0 high, 1 medium, 3 low	4	Bettina Vogel
Wheel Odometry Filter	8	2025-08-12	0 high, 3 medium, 5 low	9	Yuki Harada
Payload Scale Interface	11	2025-07-14	2 high, 3 medium, 6 low	6	Ines Duarte
Edge Gateway Agent	5	2026-03-07	1 high, 2 medium, 2 low	11	Sam Oyelaran
Release Orchestrator	16	2025-10-17	2 high, 2 medium, 12 low	9	Colin Achebe

5. Change-related incidents and restore times

Scope of impact in policy defect event INC-2025-063 was 3 sites and 28 robots, taken from the fleet records rather than from the narrative. Reading INC-2025-063 from its first entry, on 2025-05-27 robots parked below the battery reserve threshold were never offered work and the site ran short of movers. Assessed on its own record, incident INC-2025-063 shows that the customer-reported detection path is the finding, not the defect itself. In policy defect event INC-2025-063 the first signal was an operator noticed idle robots and raised a support ticket, and the reviewer treats that as reported detection. Cause analysis in Granite Peak Distribution incident INC-2025-063 concluded that the allocator filter excluded low-charge robots without checking whether a charge slot was actually available. incident INC-2025-063 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

INC-2025-063 belongs to the policy defect group and sits with Fleet Allocator. Granite Peak Distribution incident INC-2025-063 rests on moderate, resting on the team's own post-mortem and one supporting log extract evidence, which the reviewer states separately from the severity of the finding. Following policy defect event INC-2025-063 the team undertook to make the allocator consult charger availability before excluding a robot, with Marta Kellerman accountable by 2025-07-31. A contributing factor in Fleet Allocator incident INC-2025-063 was that the charge scheduler and the allocator hold separate views of charger availability. Policy defect incident INC-2025-063 shows a short mitigation and a long tail: 6 h 05 m against 12 days. policy defect event INC-2025-063 rests on a single source, and the reviewer has said so rather than implying corroboration.

Taken with the rest of the record, resource leak incident INC-2025-041 indicates that the fix is real and the regression protection is not, so recurrence remains possible. For incident INC-2025-041 the follow-through was tested rather than assumed, and landed on 2025-05-22, verified in the reservation replay suite. Traffic Manager incident INC-2025-041 classifies as a resource leak event against Traffic Manager. On 2025-03-14 the record for incident INC-2025-041 opens: a robot lost its network link mid-traverse and its corridor reservation was never released. The corrective action for INC-2025-041 is to give every corridor reservation a lease with an expiry and add a site-level queue depth alert, and 2025-05-30 is the date the team gave. Traffic Manager incident INC-2025-041 is reported here in full because a summary of it in an earlier draft of TDR-2025-21 was misread as an assurance.

What went wrong in Traffic Manager incident INC-2025-041 was that the reservation lease had no expiry, so release depended on the departing robot. Resource leak event INC-2025-041 took 1 h 52 m to stabilise and 9 days to close out. Incident INC-2025-041 reached the engineering team when a site supervisor telephoned the on-call engineer after robots queued at one intersection. Contributing to Traffic Manager incident INC-2025-041, and not repaired by the

remediation, is that the alert rule for queue depth existed but was scoped to a metric the site did not emit. Grading incident INC-2025-041 on evidence rather than on consequence gives limited; the log retention window had closed before the reviewer asked for the raw data. INC-2025-041 was checked against the company's own register, which had not been updated since the change landed.

On Remote Operations Console incident INC-2025-078 the evidence is moderate, resting on the team's own post-mortem and one supporting log extract; the severity assessment is stated separately below. The engineering team's root cause for Harborline Systems incident INC-2025-078 is that the session list cached the site roster and only invalidated it on a page reload, and the reviewer found the analysis sound. INC-2025-078 belongs to the stale state group and sits with Remote Operations Console. The trigger recorded for Harborline Systems incident INC-2025-078 on 2025-06-24 was that the console listed robots that had been removed from a site and an operator commanded a decommissioned unit. Reported detection applies to stale state incident INC-2025-078, since an internal engineer noticed the discrepancy during a demonstration. stale state incident INC-2025-078 was raised again with the owning engineer after the initial interview, and the position did not change.

Contributing to Harborline Systems incident INC-2025-078, and not repaired by the remediation, is that no test covered a roster change during an open operator session. Stale state event INC-2025-078 was contained to 4 site or tenant with no robot taken out of service. As to whether the Harborline Systems incident INC-2025-078 remediation actually landed: landed on 2025-07-28; the regression test was still marked flaky at review. Duration for incident INC-2025-078 splits into 1 h 15 m of impact and 34 days to close the underlying cause. INC-2025-078 produced one commitment: to invalidate the roster cache on a fleet membership event and add a session-level regression test, owned by Anjali Sekhon, dated 2025-08-31. INC-2025-078 was raised again with the owning engineer after the initial interview, and the position did not change.

Safety Controller incident INC-2025-090 is not read as an isolated failure: the fix is real and the regression protection is not, so recurrence remains possible. On delivery, INC-2025-090 shows this: the inspection change landed on 2025-08-04; nine units await retrofit. Detection came through the robot self-reported the fault and the safety case review picked it up, which places hardware fault event INC-2025-090 on the reported side of the ledger. On incident INC-2025-090 the evidence is moderate, resting on the team's own post-mortem and one supporting log extract; the severity assessment is stated separately below. Time to mitigate hardware fault event INC-2025-090 was immediate stop, 22 m to return to service, and time to resolve was 41 days. Fathom Mobility incident INC-2025-090 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Hardware fault event INC-2025-090 traces back to a single fact: a marginal connector on one channel produced intermittent readings inside the accepted noise band. Hardware fault incident INC-2025-090 produced one commitment: to add the connector torque check to incoming inspection and retrofit the affected harness batch, owned by Bettina Vogel, dated 2025-09-30. The trigger recorded for incident INC-2025-090 on 2025-07-19 was that a protective stop was triggered by a false positive on one safety channel while the second channel read clear. Incident INC-2025-090 affected one site and 3 robots. Contributing to incident INC-2025-090, and not repaired by the remediation, is that the connector torque specification was not part of the incoming inspection sheet. INC-2025-090 is the clearest instance in TDR-2025-21 of engineering practice being better than the documentation suggests.

So far as reach is concerned, process defect event INC-2025-146 involved no production site and no robot directly, the exposure being to internal systems. Cause analysis in Larkspur Home incident INC-2025-146 concluded that the re-run procedure required a manual ledger reset that the operator skipped. Following process defect incident INC-2025-146 the team undertook to fold the reset into the re-run command and make double-counting detectable in the ledger, with Nadia Semenova accountable by 2025-12-31. Moderate; the narrative is consistent but the reviewer could not independently reproduce it is the evidence grade for INC-2025-146, and the reader should not read severity into it. As to whether the INC-2025-146 remediation actually landed: landed on 2025-11-30. incident INC-2025-146 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Process defect event INC-2025-146 records that on 2025-11-19, a re-run of a failed metering window double-counted usage for one customer. Larkspur Home incident INC-2025-146 is not read as an isolated failure: the same class of defect can recur in two sibling components that were not changed. INC-2025-146 belongs to the process defect group and sits with Billing Usage Meter. The two durations recorded for Billing Usage Meter incident INC-2025-146 are not

applicable; caught before billing to mitigate and 11 days to resolve. Contributing to process defect incident INC-2025-146, and not repaired by the remediation, is that the runbook described the reset in prose rather than making it part of the command. Billing Usage Meter incident INC-2025-146 illustrates why TDR-2025-21 separates the severity of a finding from the confidence behind it.

Incident INC-2026-015 closure status is landed on 2026-03-03. Reported detection applies to INC-2026-015, since the on-robot supervisor raised a validation fault on the next mission. Validation event INC-2026-015 affected 2 sites and 7 robots. Underneath validation event INC-2026-015 lies the design decision that precondition validation ran at publication time rather than at apply time. Incident INC-2026-015 is not read as an isolated failure: the fix is real and the regression protection is not, so recurrence remains possible. INC-2026-015 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Mitigation of Config Distribution incident INC-2026-015 took 44 m and full resolution took 22 days. The remediation committed after validation incident INC-2026-015 was to move precondition validation to apply time and add an offline apply case to the platform suite, owned by Sam Oyelaran with a due date of 2026-03-31. Moderate, resting on the team's own post-mortem and one supporting log extract is the evidence grade for INC-2026-015, and the reader should not read severity into it. 2026-02-09 is the date of Config Distribution incident INC-2026-015, on which a robot that had been offline at publication applied a configuration bundle without re-validating preconditions. In incident INC-2026-015, the offline apply path was added for a customer with intermittent connectivity and was lightly reviewed, which is a process finding rather than a code finding. validation incident INC-2026-015 was inspected in the repository rather than described from the ticket, which is why the account is specific.

Bettina Vogel committed after safety margin incident INC-2026-033 to measure closing rate where the sensing permits and re-run the protective distance calculation, due 2026-08-31. The classification applied to INC-2026-033 is safety margin, and the affected component is Speed and Separation Monitor. Safety margin event INC-2026-033 affected one site and 6 robots. Moderate on the cause and strong on the delivery position, which was checked in the repository is the evidence grade for Asterion Logistics incident INC-2026-033, and the reader should not read severity into it. Reading safety margin incident INC-2026-033 from its first entry, on 2026-03-25 a speed reduction engaged later than expected when an operator approached faster than the assumed walking speed. safety margin event INC-2026-033 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

As to whether the incident INC-2026-033 remediation actually landed: not landed; a design study is running and the vendor assessment is booked. Drawing the evidence in incident INC-2026-033 together, the elapsed resolution time reflects a scheduling decision rather than an engineering difficulty. Safety margin event INC-2026-033 was mitigated in immediate speed reduction; 3 h to review the fleet; the resolution ran to open at the review date. A contributing factor in Asterion Logistics incident INC-2026-033 was that the assumption was documented in the internal safety case and accepted at the time. The engineering team's root cause for safety margin event INC-2026-033 is that the protective distance uses a fixed assumed approach speed rather than a measured closing rate, and the reviewer found the analysis sound. incident INC-2026-033 illustrates why TDR-2025-21 separates the severity of a finding from the confidence behind it.

1 h 04 m elapsed before INC-2026-039 was mitigated, with resolution at 19 days. Order Broker incident INC-2026-039 touched one site and 1 robots, and Crestline Fulfillment was the affected account. The reviewer treats one contributing factor in Order Broker incident INC-2026-039 as the more important lesson: the journal reconciliation runs nightly, so the orphan was invisible until the morning. Detection came through the site supervisor reported a blocked location, which places incident INC-2026-039 on the reported side of the ledger. 2026-04-07 is the date of Order Broker incident INC-2026-039, on which an order cancellation raced with assignment and left an orphaned task that blocked a pick face for an hour. incident INC-2026-039 is included because a reader of TDR-2025-21 should be able to disagree with the inference without repeating the work.

What went wrong in concurrency event INC-2026-039 was that cancellation and assignment wrote to the task record without a shared lock. Taken with the rest of the record, Crestline Fulfillment incident INC-2026-039 indicates that the commitment date was met and the verifying test now exists, which is not the pattern across the whole register. Remediation for INC-2026-039 is to serialise cancellation against assignment and add an hourly orphan sweep; Grace Lindqvist owns it and the committed date is 2026-05-31. On delivery, INC-2026-039 shows this: landed on 2026-04-26.

Incident INC-2026-039 classifies as a concurrency event against Order Broker. incident INC-2026-039 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Schedule S - Rollback and recovery drills observed or evidenced (TDR-2025-21)

Drill	Date	Component	Scope	Result	Gap identified
RD-2025-01	2025-08-30	Battery Management Service	one region	failed and was retried	restore time above target
RD-2025-02	2025-09-17	Speed and Separation Monitor	one region	failed and was retried	approval not recorded
RD-2025-03	2025-10-03	Gripper Controller	one ring	succeeded with an observation	none
RD-2025-04	2025-10-16	Fleet Health Monitor	one ring	aborted	state left in the paused ring
RD-2025-05	2025-11-02	Lidar Driver	one site	aborted	runbook step missing
RD-2025-06	2025-11-20	WMS Connector	one ring	succeeded	none
RD-2025-07	2025-12-05	Hardware-in-the-Loop Rig	one ring	succeeded	state left in the paused ring
RD-2025-08	2025-12-21	Simulation Harness	the full fleet in staging	succeeded	none
RD-2025-09	2026-01-07	Gripper Controller	one robot	aborted	runbook step missing
RD-2025-10	2026-01-23	WMS Connector	one robot	succeeded	approval not recorded
RD-2025-11	2026-02-08	Analytics Warehouse Loader	one robot	succeeded	none
RD-2025-12	2026-02-22	Gripper Controller	the full fleet in staging	succeeded	state left in the paused ring
RD-2025-13	2026-03-09	Hardware-in-the-Loop Rig	one ring	aborted	none
RD-2025-14	2026-03-25	Fleet Health Monitor	one region	aborted	none

5. Change-related incidents and restore times - part 2 of 2

A further extract of the 5. Change-related incidents and restore times appears here, taken in order from the point at which the previous part ended.

The measurement basis recorded when the 5. Change-related incidents and restore times opened is unchanged, so any figure here is comparable with any figure in the earlier parts. Nothing is restated from the earlier part.

Schedule S - Rollback and recovery drills observed or evidenced (TDR-2025-21) (continued, part 2 of 2)

Drill	Date	Component	Scope	Result	Gap identified
RD-2026-01	2026-04-13	Hardware-in-the-Loop Rig	the full fleet in staging	succeeded	runbook step missing
RD-2026-02	2026-04-27	Gripper Controller	one robot	failed and was retried	none
RD-2026-03	2026-05-13	Edge Gateway Agent	the full fleet in staging	aborted	none
RD-2026-04	2026-05-28	Video Snippet Service	the full fleet in staging	aborted	approval not recorded
RD-2026-05	2026-06-14	Event Bus	the full fleet in staging	failed and was retried	state left in the paused ring
RD-2026-06	2026-07-01	Config Distribution	one site	aborted	approval not recorded
RD-2026-07	2026-07-16	WMS Connector	one robot	aborted	restore time above target
RD-2026-08	2026-08-01	Charge Scheduler	the full fleet in staging	failed and was retried	none
RD-2026-09	2026-08-18	Map Authoring Studio	the full fleet in staging	aborted	none
RD-2026-10	2026-09-03	Map Authoring Studio	one robot	succeeded	none
RD-2026-11	2026-09-18	Charge Scheduler	one site	succeeded with an observation	approval not recorded
RD-2026-12	2026-10-05	Fleet Allocator	one ring	succeeded with an observation	none
RD-2026-13	2026-10-20	Simulation Harness	one region	succeeded	none
RD-2026-14	2026-11-05	Alarm Router	one robot	aborted	runbook step missing

6. Tooling and test-infrastructure debt

The business consequence recorded against register item TD-03 is that it no upstream fixes for a component on the safety-adjacent path. Retire the vendored geometry library in costmap fusion is entered at support item TD-03 as an

support item. Whether Costmap Fusion item TD-03 happens depends on funding, and it is partly funded, with the second phase unbudgeted. support item TD-03 was inspected in the repository rather than described from the ticket, which is why the account is specific.

2027-06-30 is the target for register item TD-09, whose status is committed. Event Bus item TD-09 is sized at 1.5 engineer-quarters, or $196,000 at the company's loaded rate. Register item TD-09 is funded and already staffed; the register status and the funding position are not the same thing. Event Bus item TD-09 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

The business consequence recorded against register item TD-13 is that it relocalisation exceeds the operator prompt timeout at sparse sites. Programme item TD-13 is funded and already staffed in the engineering plan the reviewer was shown. In the programme Localization Engine item TD-13 is behind a customer maintenance window that is not yet agreed. register item TD-13 is reported at the evidence strength available and has not been promoted to fit the narrative.

Partly funded, with the second phase unbudgeted is the funding position on register item TD-05. Sequencing of register item TD-05 matters: it is behind a platform change that has its own target date. 0.5 engineer-quarters is the estimate on TD-05 on Hardware-in-the-Loop Rig, which converts to $96,000. test capability item TD-05 was raised again with the owning engineer after the initial interview, and the position did not change.

The plan treats TD-04 on Costmap Fusion as not funded, and carried in the register regardless, which the reviewer checked against the headcount model. 2 engineer-quarters is the estimate on TD-04 on Costmap Fusion, which converts to $240,000. Raise robot software line coverage above sixty per cent on the perception path is entered at TD-04 as an quality item. TD-04 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

About TD-23 it should be said that raised by the reviewer rather than present on the company's own register. 2026-12-31 is the target for TD-23, whose status is committed. Sizing on Charge Scheduler item TD-23 gives 0.75 engineer-quarters and $92,000. TD-23 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Analytics Warehouse Loader item TD-10 carries an estimate of 0.5 engineer-quarters; the costed figure is $54,000. Programme item TD-10 carries this observation: the warehouse vendor has published a withdrawal date. What TD-10 protects against is the fact that it the analytics load fails when the function is withdrawn. support item TD-10 is the clearest instance in TDR-2025-21 of engineering practice being better than the documentation suggests.

Exposure carried by TD-21 is that it cell imbalance can hide until derating. As at the review date quality item TD-21 was in progress, with 2026-09-30 still the stated target. In the programme TD-21 on Battery Management Service is behind a platform change that has its own target date. TD-21 on Battery Management Service is the clearest instance in TDR-2025-21 of a control that exists on paper and is weaker in practice.

On TD-19 the reviewer records that the most significant open safety item in the reviewed record. The register shows register item TD-19 as in progress, due 2026-08-31. The dependency structure places safety item TD-19 behind a platform change that has its own target date. Speed and Separation Monitor item TD-19 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Validate and ship the wheel slip detector is entered at TD-20 as an safety item. TD-20 on Wheel Odometry Filter appears in the register and is funded in the current financial year. Effort on register item TD-20 is estimated at 1.5 engineer-quarters, costed at $162,000. programme item TD-20 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

On register item TD-06 the reviewer records that blocked on the physics core determinism work. TD-06 is sized at 1 engineer-quarter, or $115,000 at the company's loaded rate. TD-06 appears in the register and is not funded, and carried in the register regardless. register item TD-06 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Left in place, the condition behind TD-18 investigation evidence is lost between weekly extractions. Under safety evidence, entry TD-18 on Safety Controller is to aggregate the safety controller fault log to the cloud continuously. $104,000 is the costed estimate for programme item TD-18, based on 0.75 engineer-quarters. register item TD-18 was one of two items where TDR-2025-21 could not obtain the underlying data within the engagement.

Remote Operations Console item TD-08 is backlog with a target date of 2027-01-31. $132,000 is the costed estimate for TD-08, based on 1 engineer-quarter. Exposure carried by security supportability item TD-08 is that it no upstream security updates for a customer-facing component. register item TD-08 is the entry that changed most between the first and second document requests behind TDR-2025-21.

Register item TD-11 is sequenced behind a platform change that has its own target date. Status on configuration item TD-11 is in progress; the target is 2026-09-30. Site Provisioning Service carries configuration item TD-11, the work to re-converge the four forked site provisioning templates. configuration item TD-11 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

7. Findings in the build and release path

On assessment finding SEC-2026-04 the tester's evidence was that payloads recovered from a staging log store, and the reviewer grades it moderate. Remediated is the status of SEC-2026-04 against WMS Connector, dated 2026-05-20. 2026-04-16 is the discovery date of Medium-severity finding SEC-2026-04, which finds that wMS connector logged customer order payloads at debug level. Residual risk after remediation of finding SEC-2026-04 is low; payload logging requires an explicit per-tenant approval. SEC-2026-04 was one of two items where TDR-2025-21 could not obtain the underlying data within the engagement.

Substantiating finding SEC-2026-12 is the fact that keys recovered from a returned module, graded moderate, because the demonstration was performed in a laboratory tenant rather than in production. Reading Low-severity finding SEC-2026-12, the defect is that edge gateway agent retained mutual authentication keys after decommissioning; the discovery date is 2026-04-20. SEC-2026-12 matters because a disposed compute module carries a usable fleet credential. Partially remediated is the status of SEC-2026-12, dated pending. Edge Gateway Agent finding SEC-2026-12 is reported at the evidence strength available and has not been promoted to fit the narrative.

On Simulation Harness finding SEC-2026-06 the tester's evidence was that role policy read from the cloud configuration export, and the reviewer grades it moderate, because the demonstration was performed in a laboratory tenant rather than in production. A compromised scenario could read production artefact storage is the exposure assessment finding SEC-2026-06 describes. SEC-2026-06 against Simulation Harness is partially remediated, with a closure date of pending. Residual exposure on Medium-severity finding SEC-2026-06 reads medium; the role was scoped down but still reads one production bucket. finding SEC-2026-06 was tested rather than accepted, and the test is described in the working papers behind TDR-2025-21.

Fleet Health Monitor finding SEC-2026-09 is remediated, with a closure date of 2026-05-08. Low-severity finding SEC-2026-09 reduces to a residual risk of low. Impact of Fleet Health Monitor finding SEC-2026-09 is not theoretical; a single query can exhaust the service's memory budget. SEC-2026-09 against Fleet Health Monitor records that fleet health monitor query interface allowed unbounded result sets, discovered on 2026-04-18. SEC-2026-09 is the reason one section of TDR-2025-21 concludes narrowly rather than generally.

Identity and Authorization finding SEC-2025-01 was opened on 2025-09-16 and states that service tokens are issued with a twelve-hour lifetime. SEC-2025-01 reduces to a residual risk of low once lifetimes are cut to one hour with refresh. Remediated is the status of Identity and Authorization finding SEC-2025-01, dated 2025-11-14. The tester scored SEC-2025-01 at 7.4 and labelled it High. SEC-2025-01 against Identity and Authorization is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Billing Usage Meter finding SEC-2026-10 is remediated, with a closure date of 2026-06-05. Weighing Low-severity finding SEC-2026-10 against the compensating controls, the fix is verified and the class of defect is not, since no test now prevents its return. SEC-2026-10 was graded Low, band 4.0, by Meridian Assurance engagement MA-2026-1. SEC-2026-10 records that billing usage ledger export written to a bucket without object lock, discovered on 2026-04-19. SEC-2026-10 against Billing Usage Meter was excluded from an earlier draft of TDR-2025-21 and reinstated after the reviewer obtained the supporting record.

SEC-2025-12 against Analytics Warehouse Loader produces this judgement: the remediation is adequate and the time to remediate is not. Assessment finding SEC-2025-12 arrives as a Low finding with a 4.2 score, and the reviewer did not adjust either. As at the review date Low-severity finding SEC-2025-12 was accepted and the closure entry reads not planned. What finding SEC-2025-12 enables is this: a loader defect or compromise could alter source telemetry.

assessment finding SEC-2025-12 is reported here in full because a summary of it in an earlier draft of TDR-2025-21 was misread as an assurance.

Reading SEC-2025-04, the defect is that map authoring export bypasses the publication review; the discovery date is 2025-09-18. Severity on Medium-severity finding SEC-2025-04 is Medium, scored 5.4 by the assessor. SEC-2025-04 translates into the following exposure: an authenticated author can distribute unreviewed geometry. The reviewer declines to soften Map Authoring Studio finding SEC-2025-04: the exposure is internal rather than customer-facing, which lowers the severity without removing the finding. Medium-severity finding SEC-2025-04 is the clearest instance in TDR-2025-21 of a control that exists on paper and is weaker in practice.

Remediated is the status of SEC-2025-08, dated 2025-10-31. Severity on SEC-2025-08 against Event Bus is Medium, scored 6.0 by the assessor. Event Bus finding SEC-2025-08 is assessed as follows: the severity is right and the ownership is unclear, since two teams each believe the other holds it. The finding at finding SEC-2025-08, dated 2025-09-23, is that event bus administrative interface reachable from the application subnet. finding SEC-2025-08 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

SEC-2025-02 records that build logs printed the full authorization header, discovered on 2025-09-17. Weighing assessment finding SEC-2025-02 against the compensating controls, the underlying pattern appears in two other components that the tester did not reach. The tester scored SEC-2025-02 against Release Orchestrator at 8.1 and labelled it High. Residual risk after remediation of Release Orchestrator finding SEC-2025-02 is low; redaction is enforced in the logging library. assessment finding SEC-2025-02 would move from a concern to a verified defect if one further measurement were supplied.

8. Performance gates and their calibration

The target load run LT-2025-11 was measured against is the 0.5 per cent error-rate ceiling in the internal service documentation. The constraint surfaced by load run LT-2025-11 is no limit reached. In the series single site, steady state run LT-2025-11 is the run that changes the order rate rather than the fleet size. run LT-2025-11 was cross-checked against the engineering record supplied for TDR-2025-21, and the two agree.

Comparing LT-2025-12 with its neighbour in the series, the difference is the warehouse system endpoint used. Single site, peak hour run LT-2025-12 sits against the five-minute configuration distribution objective; whether that target is the right one is a separate question. Single site, peak hour is the scenario at single site, peak hour run LT-2025-12: 120 robots, 2,640 orders an hour, dated 2025-10-09. single site, peak hour run LT-2025-12 is the clearest instance in TDR-2025-21 of engineering practice being better than the documentation suggests.

The fifteen-minute map publication objective is the target applied to LT-2025-13. Pass with observation is what the test record gives for load run LT-2025-13. Latency in single site, peak with a charger outage run LT-2025-13 was 310 ms at p50, 870 ms at p95 and 1,690 ms at p99, with an error rate of 0.11%. run LT-2025-13 is included because a reader of TDR-2025-21 should be able to disagree with the inference without repeating the work.

The percentile set for scenario LT-2025-14 reads 290 ms, 820 ms and 1,510 ms, with errors at 0.09%. Load run LT-2025-14, taken with the other runs in the series, shows that the test environment carried a smaller dataset than production, so the database term is understated here. The constraint surfaced by 240-robot run LT-2025-14 is no limit reached. two sites, shared region run LT-2025-14 was excluded from an earlier draft of TDR-2025-21 and reinstated after the reviewer obtained the supporting record.

Load run LT-2025-15 is recorded as pass with observation. 480-robot run LT-2025-15, taken with the other runs in the series, shows that the pass verdict depends on a target that was last recalibrated in 2024. The applicable target for scenario LT-2025-15 is the five-minute configuration distribution objective, and the reviewer checked where that target came from. scenario LT-2025-15 would move from a concern to a verified defect if one further measurement were supplied.

The result recorded for LT-2025-16 is fail against the 1,200 ms p95 target, which the reviewer traced to the underlying measurement. Run LT-2025-16 returned 470 ms median, 1,680 ms at the 95th percentile and 3,410 ms at the 99th, erroring at 0.34%. On 2025-10-17 the team ran six sites, shared region run LT-2025-16, the six sites, shared region

case, at 720 robots. LT-2025-16 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

In 240-robot run LT-2025-17 the median was 620 ms and the tail reached 5,900 ms; errors ran at 1.24%. LT-2025-17 shows where the system runs out of headroom: order broker journal write queue depth exceeded its alert threshold. Whether the read replicas were enabled is what separates scenario LT-2025-17 from the adjacent run. scenario LT-2025-17 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Reproduced the November 2025 control delay before the topic split was the constraint load run LT-2025-18 exposed. The 1,200 ms p95 objective published for order assignment is the target applied to run LT-2025-18. 480 robots and 9,400 orders an hour define telemetry burst with control traffic on the shared topic run LT-2025-18, run on 2025-10-22. run LT-2025-18 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

The reviewer's reading of load run LT-2025-19 is that the pass verdict depends on a target that was last recalibrated in 2024. Set beside the other runs, 480-robot run LT-2025-19 varies only in the number of sites in the region rather than the order rate. Load run LT-2025-19 is recorded as fail against the error rate target. LT-2025-19 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

The database failover, allocator primary test, recorded as database failover, allocator primary run LT-2025-20, was executed on 2025-10-24. Weighed against the projected fleet, scenario LT-2025-20 indicates that the error rate rather than the latency is the term that would first breach a customer commitment. Errors in run LT-2025-20 were 0.96%, with latency of 340 ms, 1,410 ms and 4,200 ms across the three percentiles. LT-2025-20 was raised again with the owning engineer after the initial interview, and the position did not change.

The applicable target for load run LT-2026-01 is the recovery objective of forty seconds for a database failover, and the reviewer checked where that target came from. The constraint surfaced by 720-robot run LT-2026-01 is the control topic split removed the telemetry coupling. Measured latency in scenario LT-2026-01 was 395 ms at the median, 1,090 ms at p95 and 2,050 ms at p99. six sites after the topic split run LT-2026-01 is treated as material to the investment case rather than as a routine engineering item.

960-robot run LT-2026-02 is best read against the neighbouring run in the same series, from which it differs in one platform change between otherwise identical runs. LT-2026-02 exercised the eight sites, shared region case with 960 robots and 18,200 orders an hour, on 2026-02-12. On scenario LT-2026-02 the reviewer notes that the measurement is credible and the extrapolation the company draws from it is not. scenario LT-2026-02 is treated as material to the investment case rather than as a routine engineering item.

960-robot run LT-2026-03 shows where the system runs out of headroom: read offload bought headroom without a schema change. Run LT-2026-03 is scored against the 1,500 ms p95 acknowledgement objective for the warehouse connector, which the company set in its own service documentation. The test bed for scenario LT-2026-03 carries a replayed order stream taken from the two largest sites, which is the main reason the reviewer does not treat it as production-equivalent. scenario LT-2026-03 was raised again with the owning engineer after the initial interview, and the position did not change.

Run LT-2026-04 used the standard performance environment, which is characterised by the production cloud configuration and a laboratory network profile. Comparing LT-2026-04 with its neighbour in the series, the difference is the number of sites in the region rather than the order rate. The percentile set for run LT-2026-04 reads 520 ms, 1,720 ms and 4,050 ms, with errors at 0.48%. LT-2026-04 is treated as material to the investment case rather than as a routine engineering item.

The reviewer's reading of order burst after the broker queue change run LT-2026-05 is that the pass verdict depends on a target that was last recalibrated in 2024. Order burst after the broker queue change run LT-2026-05 was executed in the performance environment, which carries a synthetic robot fleet driven from recorded mission traces. Pass with observation is what the test record gives for run LT-2026-05. order burst after the broker queue change run LT-2026-05 was excluded from an earlier draft of TDR-2025-21 and reinstated after the reviewer obtained the supporting record.

9. How a release reaches the sites

The reason Reno, Nevada appears here is that it is the smallest customer deployment. Reno, Nevada runs 29 robots for Blue Mesa Components and opened in 2025-05. Reno, Nevada was checked against the company's own register, which had not been updated since the change landed.

Vendor B is the warehouse system at Tracy, California, where headroom is 36%. Tracy, California is notable in this review because it is the site whose cleaning schedule now overlaps the peak shift. Tracy, California is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Against its peak profile, Newnan, Georgia retains 33% headroom on a vendor B integration. 2025-07 is the go-live month at Newnan, Georgia, where Ironwood Grocery Group now runs 66 robots. Newnan, Georgia is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Remaining capacity at Mesquite, Texas is 71%, measured against the site's peak order profile. The Mesquite, Texas deployment serves Evergreen Warehousing with 16 robots, live since 2026-06. Mesquite, Texas is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Remaining capacity at Wilmington, Ohio is 14%, measured against the site's peak order profile. 112 robots are deployed at Wilmington, Ohio, which went live in 2023-08. Wilmington, Ohio was inspected in the repository rather than described from the ticket, which is why the account is specific.

Lockbourne, Ohio came online in 2026-04 and now carries 19 robots for Granite Peak Distribution. What distinguishes Lockbourne, Ohio is that it is commissioned after the template baseline check landed. Lockbourne, Ohio would be a candidate for the first post-close engineering review under any reasonable plan.

The Shepherdsville, Kentucky deployment serves Fathom Mobility with 47 robots, live since 2025-03. Vendor C is the warehouse system at Shepherdsville, Kentucky, where headroom is 41%. Shepherdsville, Kentucky was checked against the corresponding entry in the company's own register, and the register was current.

Vendor C is the warehouse system at Ontario, California, where headroom is 26%. For Ontario, California the material fact is that it is the site whose shift pattern change caused the April 2025 charge incident. Ontario, California is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Meridian, Idaho is notable in this review because it is the longest-running customer deployment and the source of the most incident history. The Meridian, Idaho site integrates with vendor A and shows 22% headroom. Meridian, Idaho was reviewed against the position twelve months earlier, and nothing measurable has changed.

Site ST-01 is Fremont, California: internal engineering site, 34 robots, live from 2021-06. The Fremont, California site integrates with test harness and shows not measured headroom. Fremont, California sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

78 robots are deployed at Grand Prairie, Texas, which went live in 2023-04. Grand Prairie, Texas is the second Asterion site, commissioned on the same template, which is why the reviewer sampled it. Grand Prairie, Texas would move from a concern to a verified defect if one further measurement were supplied.

Kestrel Medical Supply operates 37 Helioscope robots at Bethlehem, Pennsylvania, commissioned in 2025-11. On Bethlehem, Pennsylvania the record shows that it is sparse survey features, which is why relocalisation is slow here. Bethlehem, Pennsylvania was reviewed against the position twelve months earlier, and the direction of travel is favourable.

The Edwardsville, Illinois deployment serves Crestline Fulfillment with 24 robots, live since 2026-03. On Edwardsville, Illinois the record shows that it is the newest Crestline site, still ramping. Edwardsville, Illinois is reported at the evidence strength available and has not been promoted to fit the narrative.

Lehigh Valley, Pennsylvania is notable in this review because it is commissioned during the connector rework. Against its peak profile, Lehigh Valley, Pennsylvania retains 38% headroom on a vendor B integration. Lehigh Valley, Pennsylvania is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Chino, California is the go-live delayed by the forked provisioning template, which is why the reviewer sampled it. 2026-01 is the go-live month at Chino, California, where Larkspur Home now runs 33 robots. Chino, California was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

10. Process findings management does not accept

The present report keeps declined items visible: risk acceptance without an expiry date is itself recorded as a finding. Where management has declined to act, this report preserves the decision and the stated rationale so a post-close plan can revisit it. Brandywine Technology Partners notes that 2 of the matters in report TDR-2025-21 are capable of being closed before completion if the company is asked in time. 7 items in report TDR-2025-21 are post-close engineering work rather than transaction issues, and the reviewer has costed them at $1,405,000. S. Alderton has stated the limitations at the point they bite rather than collecting them into a disclaimer.

The report at TDR-2025-21 closes with 4 open items, listed with owner, evidence and the reviewer's view of whether the committed date is realistic. 4 matters remain open at the date of this report, and none of them is closed by the passage of time alone. Where TDR-2025-21 says a defect is verified, the reviewer reproduced it or read a production trace of it; everything else is recorded as a concern. Nothing in this report is presented as verified on the strength of an interview alone. S. Alderton has not smoothed the sequence of this report; it follows the order in which the evidence arrived.

Report TDR-2025-21 distinguishes three positions: verified defect, credible concern, and matter requiring further work. The third category is not empty. The evidence convention in the present report is that severity describes consequence and evidence strength describes confidence, and the two are never merged. The present report covers 13 of 34 services; the untested remainder is listed so a reader can see what was not reached. This report added two components to its sample after the first week, because the interview evidence pointed at them. Brandywine Technology Partners has reported the position as at 2025-11-06 and has not attempted to predict where it will be at completion.

TDR-2025-21 proposes 5 metrics for the post-close monitoring pack, each already produced by an existing system. S. Alderton notes in this report that the company's own reporting is adequate for three of the four recommended indicators. Work on report TDR-2025-21 ran over 5 weeks and included 20 interviews and read-only access to the engineering repositories. The working method behind TDR-2025-21 was to start from the incident record and work back to the design, rather than the other way round. The numbering in the report at TDR-2025-21 follows the order of the work rather than the order of importance.

11. Conclusions on delivery capability

For Ardent Peak's purposes, this report concludes that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. The report at TDR-2025-21 concludes that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. Report TDR-2025-21 preserves 4 unresolved questions rather than resolving them by assumption. The reviewer has resisted the temptation to tidy the present report: the open items are open because the evidence is not there yet. Figures quoted in the present report come from the company's systems unless the text says the reviewer measured them.

Deal implications in this report are modest in aggregate and specific in detail: 2 items for the closing checklist, 7 for the plan. Nothing in this report supports a retention or an escrow; the findings are operational and they are funded. As instructed, Brandywine Technology Partners confined the report at TDR-2025-21 to the engineering delivery process, pipeline metrics and change-management controls; three adjacent questions raised during the work are listed as follow-ups rather than answered here. Brandywine Technology Partners delivered this report on 2025-11-06; the scope agreed with Ardent Peak was the engineering delivery process, pipeline metrics and change-management controls. Brandywine Technology Partners has reported unresolved disagreements as disagreements rather than choosing between the accounts.

5 indicators are recommended in the report at TDR-2025-21; the reviewer deliberately avoided proposing anything that needs a new pipeline. The report at TDR-2025-21 would have the monitoring pack include declined items with their compensating controls, so that acceptance decisions age visibly. Sampling in the report at TDR-2025-21 covered 13 of the 34 catalogued services and 5 of the 18 live sites. The population for report TDR-2025-21 was the service inventory as at

2025-11-06; 13 entries were examined and the rest were reviewed only at inventory level. S. Alderton has recorded the date of each measurement, because several of them will be stale within a quarter.

Test coverage on Edge Gateway Agent is 74 per cent line coverage and the measurement basis was confirmed with Sam Oyelaran. Edge Gateway Agent was assessed against the objective and the defect record, and the risk is concentrated in one code path that has neither automated coverage nor a bench equivalent. Edge Gateway Agent has a single named owner, Sam Oyelaran of Platform Infrastructure, which is not true of every service reviewed. The objective recorded against Edge Gateway Agent is that no telemetry is lost for an outage shorter than four hours, and it is reported weekly to the engineering leadership. For Edge Gateway Agent the supportability position is none; the transport library was updated in early 2026. Edge Gateway Agent is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Against Charge Scheduler, the coverage figure is 63 per cent line coverage, which the figure is respectable but excludes the fallback path, which is where the known defect sits. Underlying Charge Scheduler in this report are the scheduler policy document and four sites' charge utilisation exports. Nadia Semenova of Fleet Services maintains Charge Scheduler, and the service inventory entry was current at the review date. Charge Scheduler publishes a stable interface, though two fields used by the fleet allocator and the fleet health monitor are documented only in code comments. The cadence recorded for Charge Scheduler is fortnightly, without a formal approval gate until March 2026, which matches the observed release history. Charge Scheduler was tested rather than accepted, and the test is described in the working papers behind TDR-2025-21.